

TidyTuesday Week 37

Pranay Gundam

Friday 11th September, 2026

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1 Weekly Summary

AI-Generated Content

The summary below was written by Claude (Anthropic) from that week's regression outputs and series descriptions. It has not been reviewed or fact-checked by a human, and should be read as an automated, informal recap – not analysis.

Welcome back to another thrilling installment of Numbers That Have No Business Being Near Each Other. This week the random-pair generator really outdid itself, kicking off Monday by asking whether Lithuania's "Discussion About Pandemics Index" – yes, that is a real series, and yes, its popularity score is a heartbreaking 1 – could predict how much the U.S. government borrows from the public. The raw regression said "sure, a little," with an R-squared around 0.20, and here is the genuinely surprising twist: when we strip out the shared drift, the relationship barely flinches. It holds up in the detrended version too. Do I believe that Lithuanian pandemic chatter governs American public borrowing? No. Absolutely not. But it survived the one test I built specifically to embarrass it, so I am obligated to raise a single eyebrow.

Tuesday paired U.S. coal CO2 emissions with a medical price index for congenital anomalies, because of course it did, and this was the star of the week. The raw regression posted a whopping 0.61 R-squared, and instead of collapsing like a folding chair once detrended, it actually stayed strong at 0.56 – and the coefficient roughly doubled, which is the statistical equivalent of a series flexing at you. I have no earthly theory here beyond "two things happened over the same twenty years," but as trend-mirages go, this one refuses to evaporate, which puts it in the rare and unsettling "actually kind of interesting" bin. Please do not build policy on coal-fueled anomaly pricing.

Then came the classic dunk we all know and love. Wednesday matched the Market Value of Gross Federal Debt against business equipment leases, and the raw regression breathlessly reported a screamingly significant p-value with a laughable R-squared of 0.09 – lots of confidence, almost no explanatory power, the dream. Amusingly the detrended version was slightly stronger, so even this one didn't fully faceplant. Thursday, meanwhile, delivered the week's most beautiful faceplant: construction quits versus women's work-from-home rate had an R-squared of basically 0.0005 raw, a p-value of 0.86, the numeric shrug of the century – and then detrending resurrected it into a significant 0.30 relationship out of nowhere. Truly a series that means nothing until you make it mean something. Friday capped things off with national wages versus manufacturing employment in Ogden, Utah, which posted a gaudy 0.80 raw R-squared that promptly deflated to 0.40 once the shared upward march of time was removed – respectable, but exactly the kind of half-mirage this project exists to poke at.

As always, the mandatory reminder: these are randomly paired economic series, glued together by a script with no supervision and even less shame. Nothing here implies causation, correlation-with-meaning, or anything a responsible person would cite. Spurious is the baseline

expectation, not the exception, and the only reason we detrend at all is so I can tell you which coincidences at least tried a little harder than the others. See you next week for more numbers pretending to know each other.

2 Date: 2026-09-07

Series ID: WPDILTU

This series is titled Discussion About Pandemics Index for Lithuania and has a frequency of Quarterly. The units are Index and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1996-01-01 and the observation end date is 2026-04-01. The popularity of this series is 1.

Series ID: MTSMFBP133FMS

This series is titled Means of Financing: Borrowing from the Public and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2010-01-01 and the observation end date is 2026-07-01. The popularity of this series is 14.

2.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_MTSMFBP133FMS	R-squared:	0.199			
Model:	OLS	Adj. R-squared:	0.186			
Method:	Least Squares	F-statistic:	15.87			
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000176			
Time:	16:43:46	Log-Likelihood:	-893.46			
No. Observations:	66	AIC:	1791.			
Df Residuals:	64	BIC:	1795.			
Df Model:	1					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	4.886e+04	2.46e+04	1.985	0.051	-321.644	9.8e+04
value_fred_WPDILTU	1205.6312	302.613	3.984	0.000	601.092	1810.170
Omnibus:	51.230	Durbin-Watson:	2.093			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	275.412			
Skew:	2.139	Prob(JB):	1.57e-60			
Kurtosis:	12.047	Cond. No.	87.5			

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

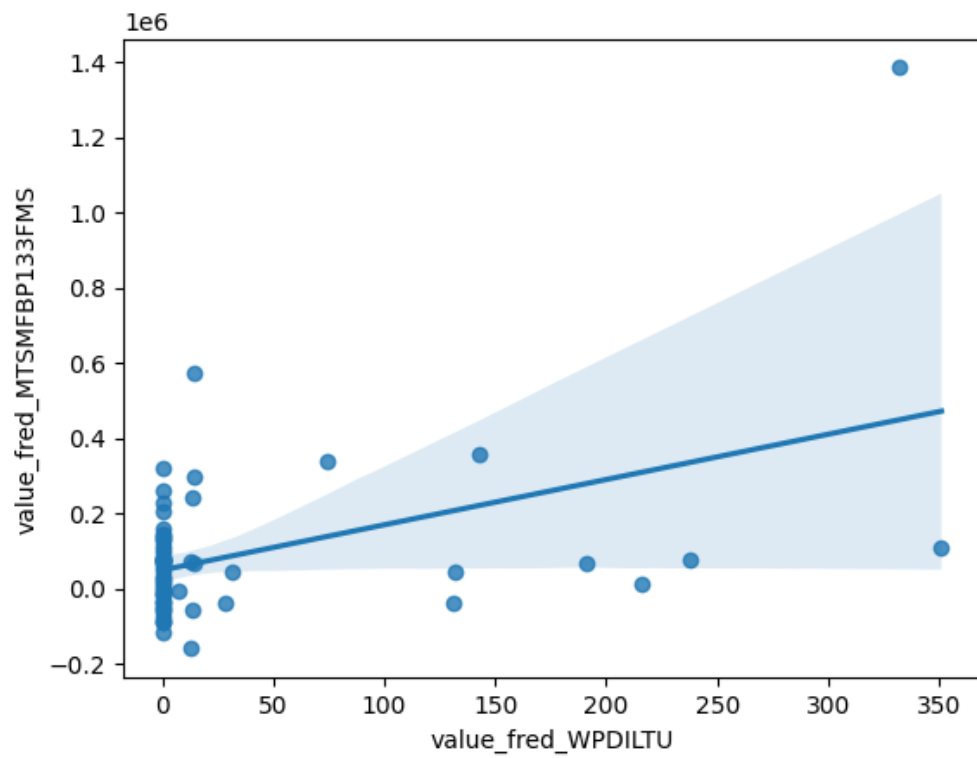


Figure 1: Raw Regression Plot for 2026-09-07

2.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_MTSMFBP133FMS_detrended	R-squared:	0.174
Model:	OLS	Adj. R-squared:	0.161
Method:	Least Squares	F-statistic:	13.48
Date:	Mon, 07 Sep 2026	Prob (F-statistic):	0.000493
Time:	16:43:46	Log-Likelihood:	-893.06
No. Observations:	66	AIC:	1790.
Df Residuals:	64	BIC:	1794.
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	5.051e-09	2.28e+04	2.22e-13	1.000	-4.55e+04	4.55e+04
value_fred_WPDILTU_detrended	1138.7087	310.105	3.672	0.000	519.204	1758.214

Omnibus:	52.012	Durbin-Watson:	2.129
Prob(Omnibus):	0.000	Jarque-Bera (JB):	302.597
Skew:	2.138	Prob(JB):	1.96e-66
Kurtosis:	12.579	Cond. No.	73.4

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

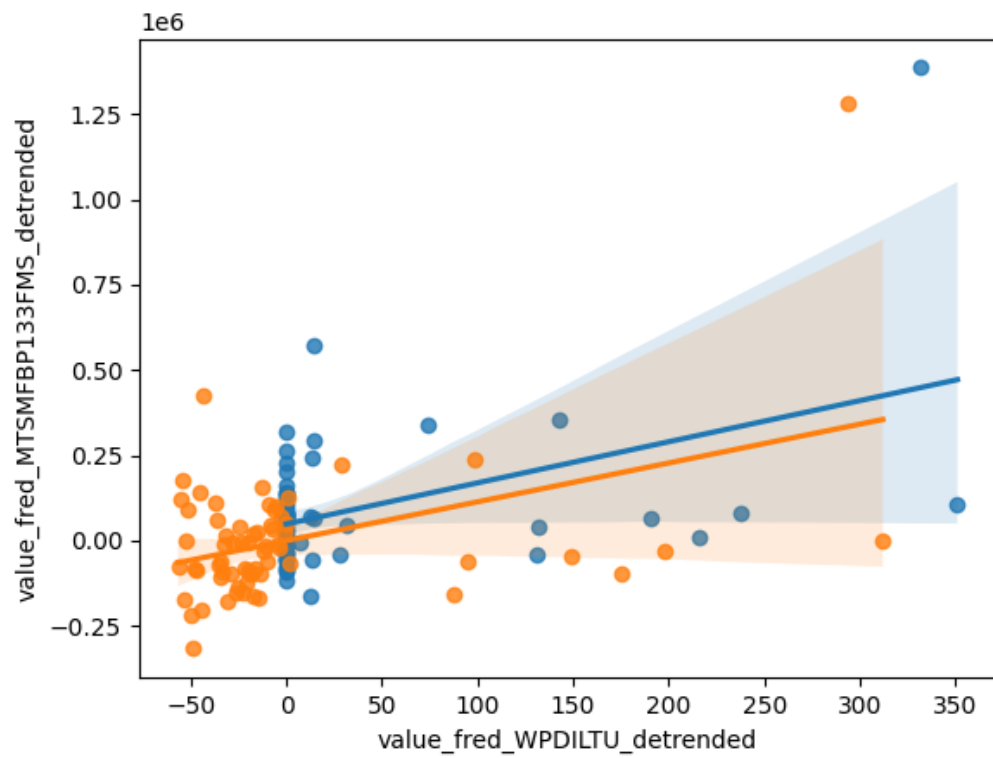


Figure 2: Detrended Regression Plot for 2026-09-07

3 Date: 2026-09-08

Series ID: EMISSCO2TOTVTTCOUSA

This series is titled Total Carbon Dioxide Emissions From All Sectors, Coal for United States and has a frequency of Annual. The units are Million Metric Tons CO2 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1970-01-01 and the observation end date is 2021-01-01. The popularity of this series is 5.

Series ID: COGANOPIMEPS

This series is titled Medical Services Expenditures by Disease: Other: Congenital Anomalies Price Index, MEPS Account Basis and has a frequency of Annual. The units are Index 2009=100 and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2000-01-01 and the observation end date is 2020-01-01. The popularity of this series is 1.

3.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_COGANOPIMEPS	R-squared:	0.606
Model:	OLS	Adj. R-squared:	0.585
Method:	Least Squares	F-statistic:	29.22
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	3.24e-05
Time:	15:27:44	Log-Likelihood:	-111.42
No. Observations:	21	AIC:	226.8
Df Residuals:	19	BIC:	228.9
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-146.4574	51.777	-2.829	0.011	-254.827	-38.088
value_fred_EMISSCO2TOTVTTCOUSA	0.1533	0.028	5.406	0.000	0.094	0.213

Omnibus:	3.022	Durbin-Watson:	1.167
Prob(Omnibus):	0.221	Jarque-Bera (JB):	1.530
Skew:	0.625	Prob(JB):	0.465
Kurtosis:	3.429	Cond. No.	8.45e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 8.45e+03. This might indicate that there are strong multicollinearity or other numerical problems.

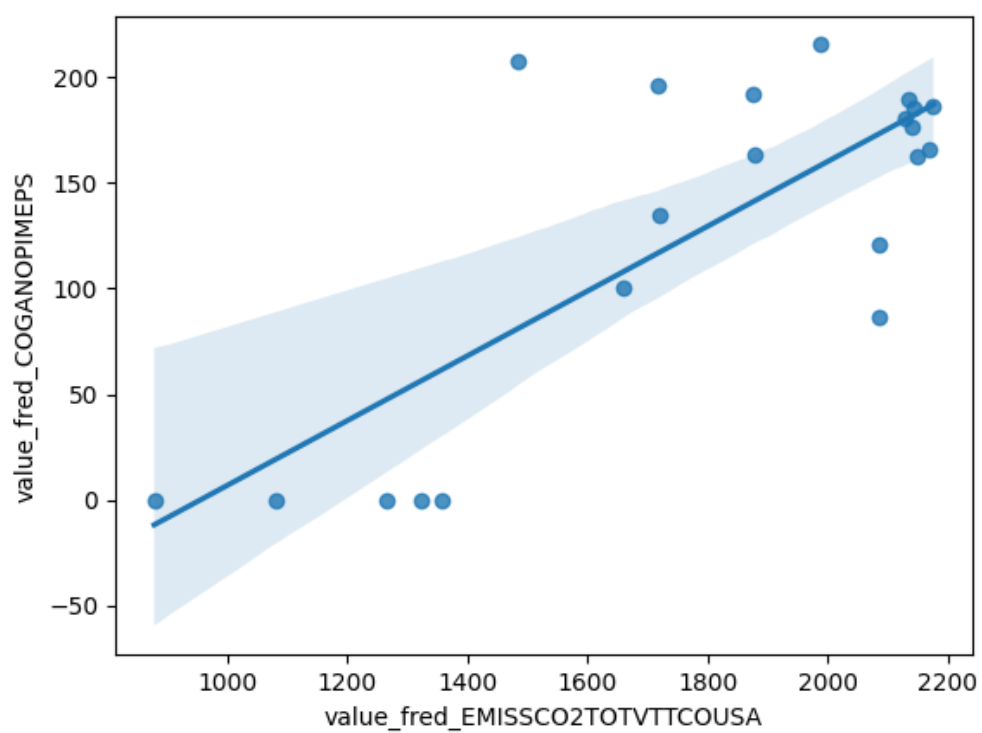


Figure 3: Raw Regression Plot for 2026-09-08

3.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_COGANOPIMEPS_detrended	R-squared:	0.559
Model:	OLS	Adj. R-squared:	0.536
Method:	Least Squares	F-statistic:	24.09
Date:	Tue, 08 Sep 2026	Prob (F-statistic):	9.76e-05
Time:	15:27:44	Log-Likelihood:	-108.14
No. Observations:	21	AIC:	220.3
Df Residuals:	19	BIC:	222.4
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-6.734e-12	9.567	-7.04e-13	1.000	-20.023	20.023
value_fred_EMISSCO2TOTVTTCOUSA_detrended	0.3032	0.062	4.908	0.000	0.174	0.432

Omnibus:	4.692	Durbin-Watson:	1.541
Prob(Omnibus):	0.096	Jarque-Bera (JB):	2.578
Skew:	0.726	Prob(JB):	0.276
Kurtosis:	3.915	Cond. No.	155.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

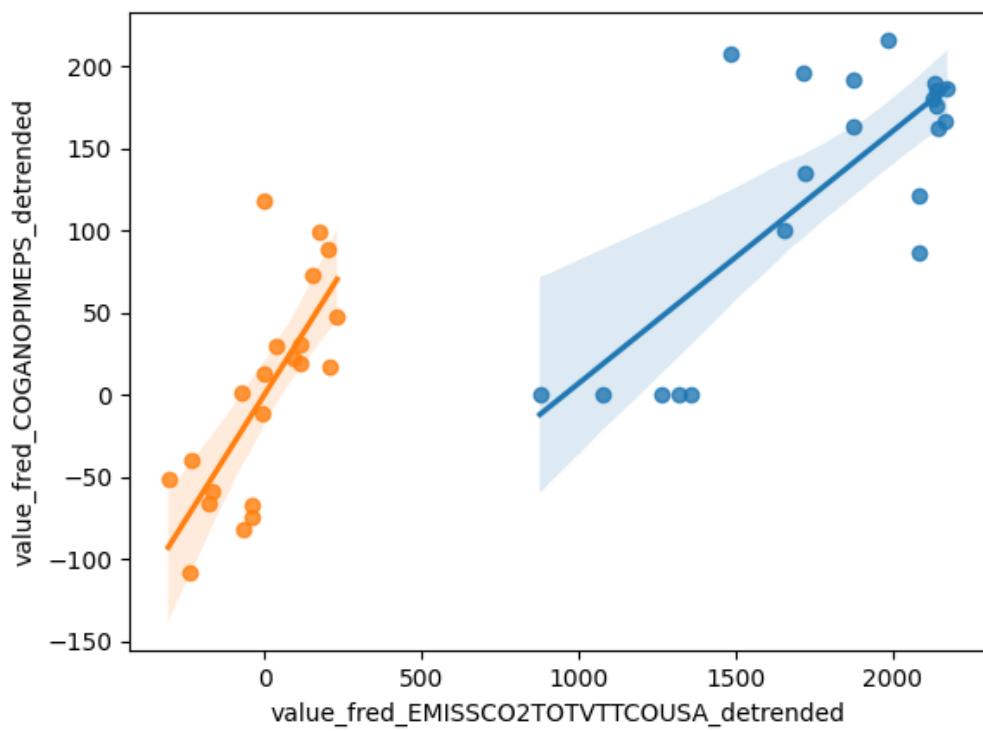


Figure 4: Detrended Regression Plot for 2026-09-08

4 Date: 2026-09-09

Series ID: MVGFD027MNFRBDAL

This series is titled Market Value of Gross Federal Debt and has a frequency of Monthly. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1942-01-01 and the observation end date is 2026-07-01. The popularity of this series is 37.

Series ID: DTBTERNM

This series is titled Business Equipment Leases Owned and Securitized by Finance Companies, Level and has a frequency of Monthly. The units are Millions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-06-01 and the observation end date is 2026-06-01. The popularity of this series is 1.

4.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_DTBTERNM	R-squared:	0.095
Model:	OLS	Adj. R-squared:	0.093
Method:	Least Squares	F-statistic:	45.33
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	5.34e-11
Time:	15:24:33	Log-Likelihood:	-5318.4
No. Observations:	433	AIC:	1.064e+04
Df Residuals:	431	BIC:	1.065e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.633e+05	4472.957	36.517	0.000	1.55e+05	1.72e+05
value_fred_MVGFD027MNFRBDAL	-1.7290	0.257	-6.732	0.000	-2.234	-1.224

Omnibus:	2749.157	Durbin-Watson:	0.021
Prob(Omnibus):	0.000	Jarque-Bera (JB):	48.079
Skew:	0.054	Prob(JB):	3.63e-11
Kurtosis:	1.371	Cond. No.	3.10e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 3.1e+04. This might indicate that there are strong multicollinearity or other numerical problems.

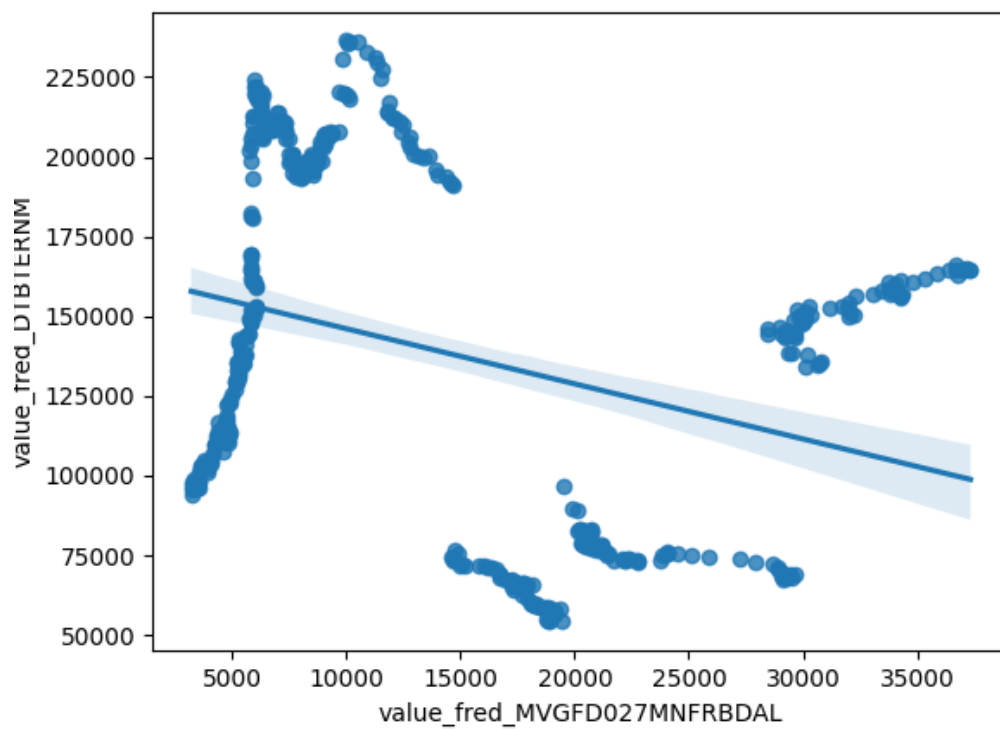


Figure 5: Raw Regression Plot for 2026-09-09

4.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_DTBTERNM_detrended	R-squared:	0.116
Model:	OLS	Adj. R-squared:	0.114
Method:	Least Squares	F-statistic:	56.72
Date:	Wed, 09 Sep 2026	Prob (F-statistic):	2.97e-13
Time:	15:24:33	Log-Likelihood:	-5302.8
No. Observations:	433	AIC:	1.061e+04
Df Residuals:	431	BIC:	1.062e+04
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975
const	9.044e-09	2427.545	3.73e-12	1.000	-4771.300	4771.300
value_fred_MVGFD027MNFRBDAL_detrended	-6.1487	0.816	-7.531	0.000	-7.753	-4.543

Omnibus:	698.313	Durbin-Watson:	0.023
Prob(Omnibus):	0.000	Jarque-Bera (JB):	33.214
Skew:	-0.174	Prob(JB):	6.13e-08
Kurtosis:	1.689	Cond. No.	2.97e+03

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 2.97e+03. This might indicate that there are strong multicollinearity or other numerical problems.

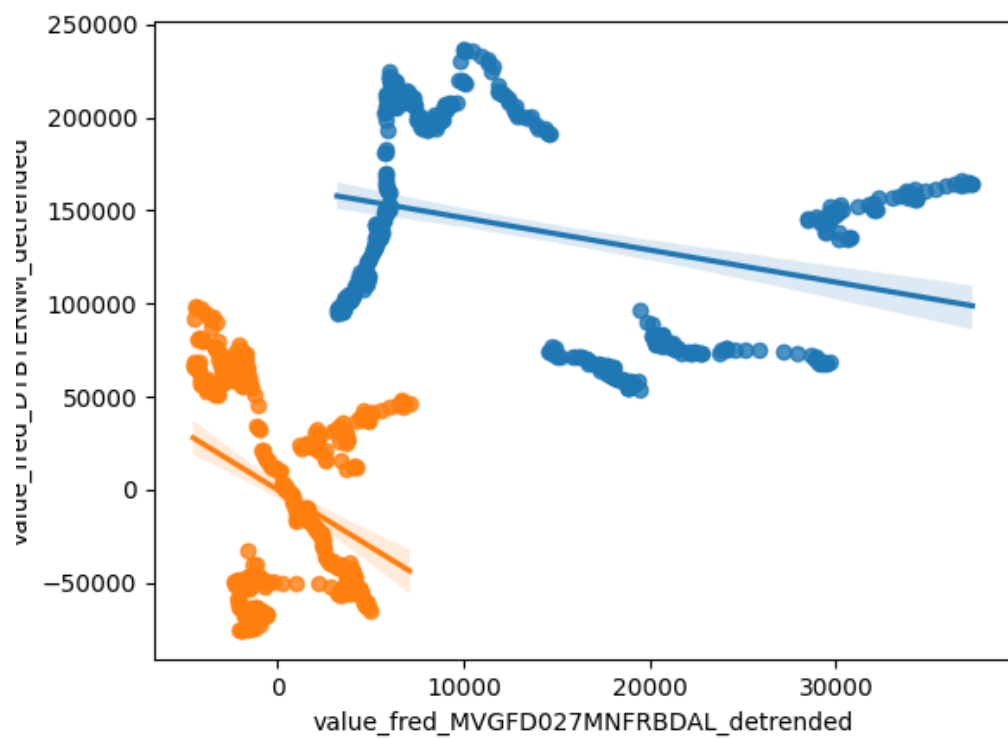


Figure 6: Detrended Regression Plot for 2026-09-09

5 Date: 2026-09-10

Series ID: JTS2300QUR

This series is titled Quits: Construction and has a frequency of Monthly. The units are Rate and the seasonal adjustment is Seasonally Adjusted. The observation start date is 2000-12-01 and the observation end date is 2026-07-01. The popularity of this series is 13.

Series ID: WFHCOVIDFRACMATWOMEN

This series is titled Work from Home Rate: Women and has a frequency of Monthly. The units are Percent of Full Paid Working Days and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 2020-05-01 and the observation end date is 2026-08-01. The popularity of this series is 6.

5.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN	R-squared:	0.000
Model:	OLS	Adj. R-squared:	-0.013
Method:	Least Squares	F-statistic:	0.03323
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	0.856
Time:	15:17:47	Log-Likelihood:	-240.33
No. Observations:	74	AIC:	484.7
Df Residuals:	72	BIC:	489.3
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	32.4960	3.704	8.774	0.000	25.113	39.879
value_fred_JTS2300QUR	-0.3289	1.804	-0.182	0.856	-3.926	3.268

Omnibus:	71.083	Durbin-Watson:	0.194
Prob(Omnibus):	0.000	Jarque-Bera (JB):	449.811
Skew:	2.992	Prob(JB):	2.11e-98
Kurtosis:	13.492	Cond. No.	12.7

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

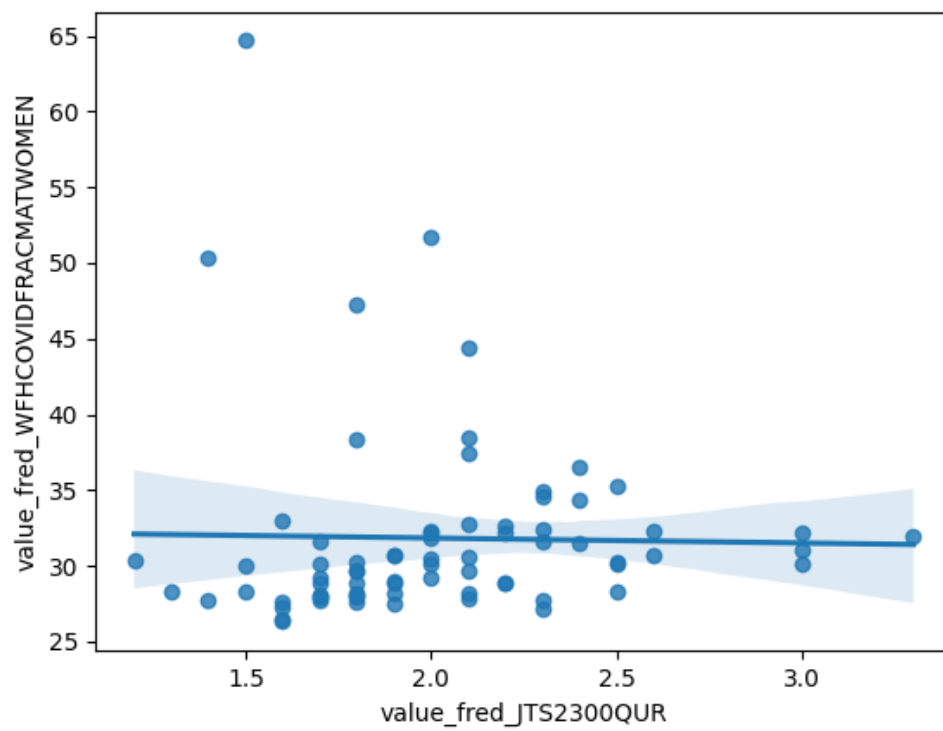


Figure 7: Raw Regression Plot for 2026-09-10

5.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_WFHCOVIDFRACMATWOMEN_detrended	R-squared:	0.300
Model:	OLS	Adj. R-squared:	0.290
Method:	Least Squares	F-statistic:	30.84
Date:	Thu, 10 Sep 2026	Prob (F-statistic):	4.44e-07
Time:	15:17:47	Log-Likelihood:	-205.78
No. Observations:	74	AIC:	415.6
Df Residuals:	72	BIC:	420.2
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-7.578e-12	0.460	-1.65e-11	1.000	-0.917	0.917
value_fred_JTS2300QUR_detrended	-7.2666	1.309	-5.553	0.000	-9.875	-4.658

Omnibus:	45.009	Durbin-Watson:	0.681
Prob(Omnibus):	0.000	Jarque-Bera (JB):	171.162
Skew:	1.828	Prob(JB):	6.80e-38
Kurtosis:	9.492	Cond. No.	2.84

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

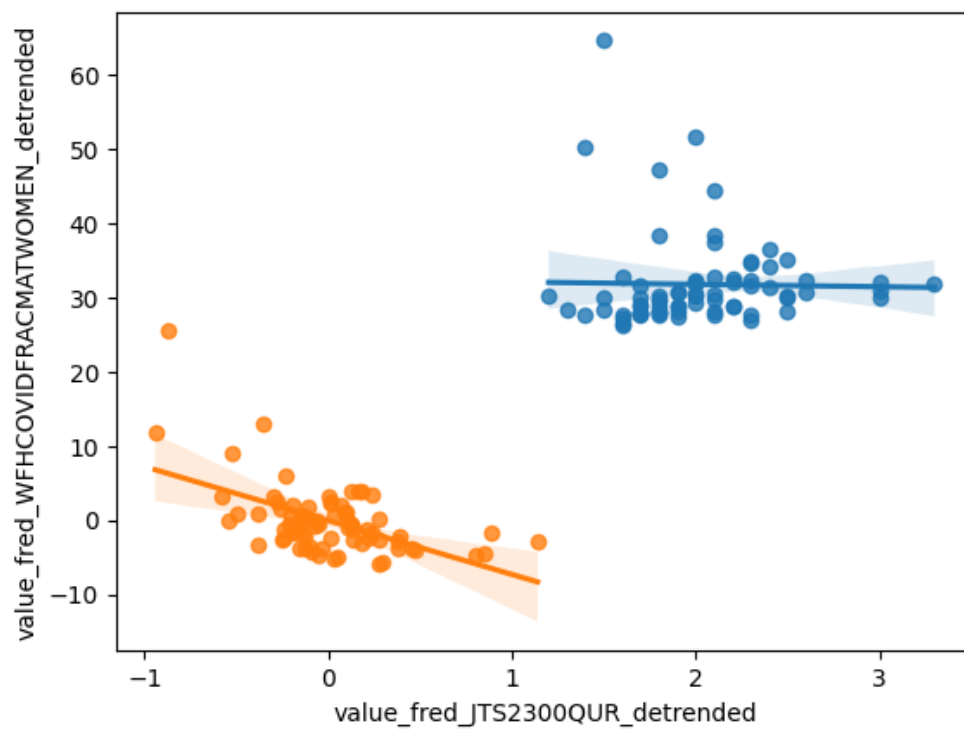


Figure 8: Detrended Regression Plot for 2026-09-10

6 Date: 2026-09-11

Series ID: A034RC1A027NBEA

This series is titled National income: Compensation of employees: Wages and salaries and has a frequency of Annual. The units are Billions of Dollars and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1929-01-01 and the observation end date is 2025-01-01. The popularity of this series is 23.

Series ID: OGDE249MFGN

This series is titled All Employees: Manufacturing in Ogden, UT (MSA) and has a frequency of Monthly. The units are Thousands of Persons and the seasonal adjustment is Not Seasonally Adjusted. The observation start date is 1990-01-01 and the observation end date is 2026-07-01. The popularity of this series is 1.

6.1 Raw Regression

Regresses the two series against each other directly. A significant result here can come just as easily from both series sharing a long-run trend as from any real relationship.

Dep. Variable:	value_fred_OGDE249MFGN	R-squared:	0.802
Model:	OLS	Adj. R-squared:	0.797
Method:	Least Squares	F-statistic:	138.0
Date:	Fri, 11 Sep 2026	Prob (F-statistic):	1.62e-13
Time:	15:19:38	Log-Likelihood:	-76.434
No. Observations:	36	AIC:	156.9
Df Residuals:	34	BIC:	160.0
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	14.0195	0.885	15.840	0.000	12.221	15.818
value_fred_A034RC1A027NBEA	0.0015	0.000	11.748	0.000	0.001	0.002

Omnibus:	2.395	Durbin-Watson:	0.177
Prob(Omnibus):	0.302	Jarque-Bera (JB):	1.478
Skew:	0.221	Prob(JB):	0.478
Kurtosis:	2.111	Cond. No.	1.82e+04

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

[2] The condition number is large, 1.82e+04. This might indicate that there are strong multicollinearity or other numerical problems.

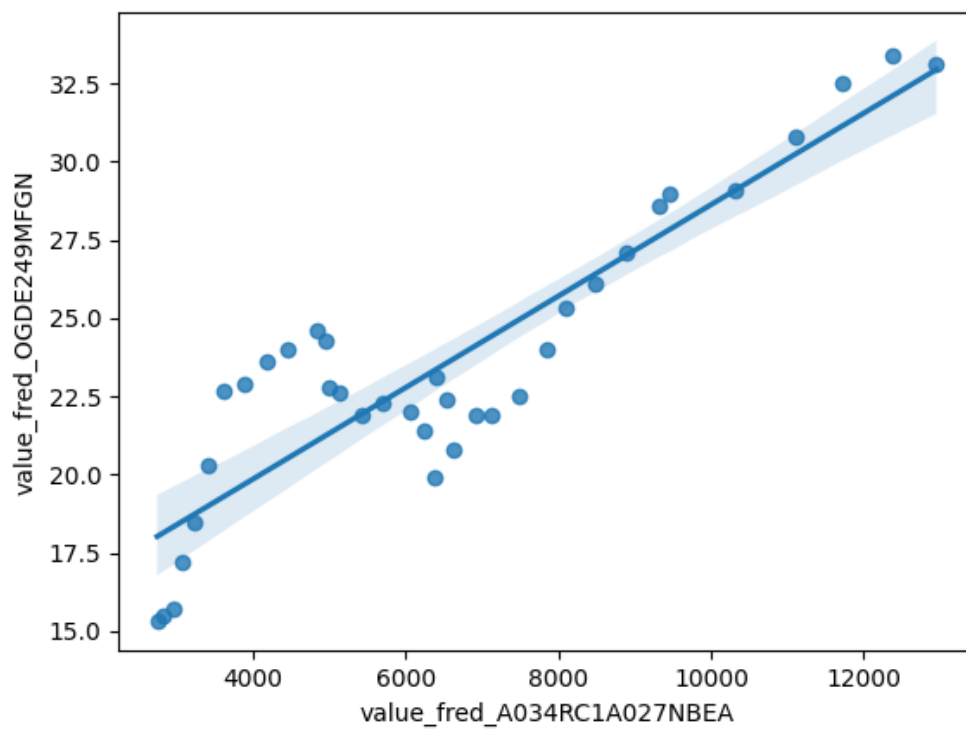


Figure 9: Raw Regression Plot for 2026-09-11

6.2 Detrended Regression

Regresses the residuals of each series after removing its own linear time trend, so a shared trend can no longer drive the result on its own. A relationship that stays significant here is better evidence of a genuine link between the two series.

Dep. Variable:	value_fred_OGDE249MFGN_detrended	R-squared:	0.401
Model:	OLS	Adj. R-squared:	0.383
Method:	Least Squares	F-statistic:	22.75
Date:	Fri, 11 Sep 2026	Prob (F-statistic):	3.40e-05
Time:	15:19:38	Log-Likelihood:	-74.459
No. Observations:	36	AIC:	152.9
Df Residuals:	34	BIC:	156.1
Df Model:	1		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	-5.583e-14	0.328	-1.7e-13	1.000	-0.667	0.667
value_fred_A034RC1A027NBEA_detrended	0.0025	0.001	4.770	0.000	0.001	0.004

Omnibus:	0.572	Durbin-Watson:	0.205
Prob(Omnibus):	0.751	Jarque-Bera (JB):	0.649
Skew:	0.067	Prob(JB):	0.723
Kurtosis:	2.356	Cond. No.	638.

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

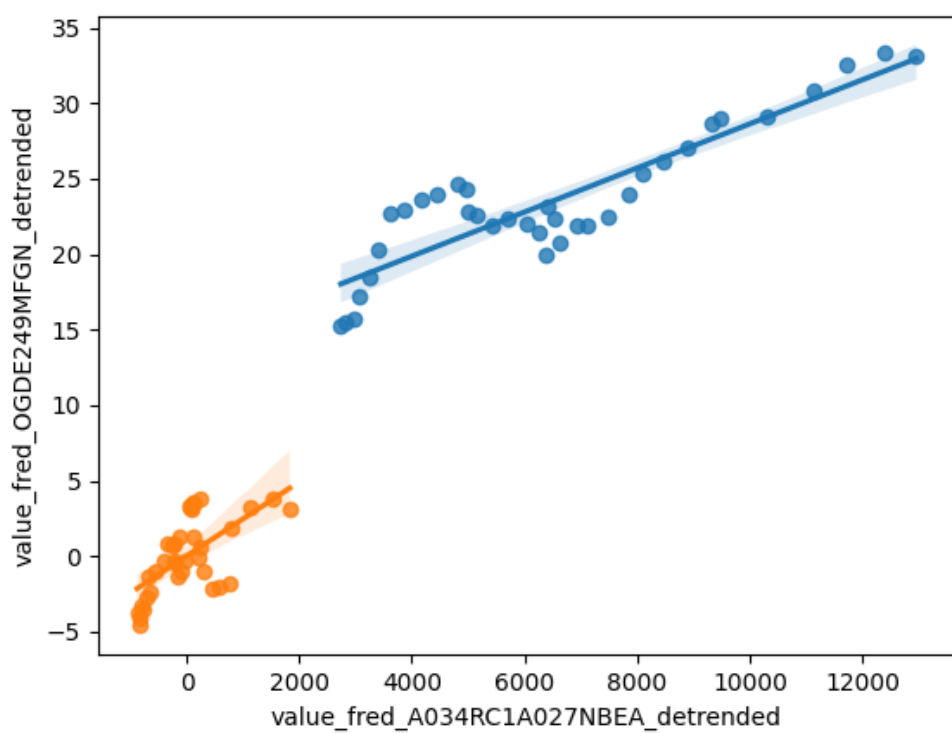


Figure 10: Detrended Regression Plot for 2026-09-11